

Building Our Digital World: Computer Systems & Architecture

COMP1860

School of Computer Science, University of Leeds

Activity Sheet 1.1

This worksheet contains a combination of formative activities and activities that contribute towards your portfolio. Portfolio activities are detailed in the section “Portfolio Task”. Activities marked by (*) are advanced activities as may take some time to complete.

Expectations:

- 1. Timeliness** You should complete all of the activities in the order provided. Activities marked as part of the portfolio should be submitted to Gradescope before the completion date (Friday, 09/10/2026, 1400).
- 2. Presentation:** You should present all of your work clearly and concisely following any additional guidance provided by the module staff in the module handbook.
- 3. Integrity:** You are responsible that the evidence you submit as part of your portfolio evidence is entirely your own work. You can find out more about Academic integrity on the Skill@library website. All work you submit for assessment is subject to the academic integrity policy.

Feedback: Feedback on formative activities will be provided via Lab classes and tutorials. Feedback on activities submitted as part of the portfolio will be available on Gradescope.

Support opportunities: Support with the activity sheet is available in the Lab classes and tutorials. Individual support is available via the online booking system.

Expected time for completion: 2-3 hours.

Expected complete date: Friday, 09/10/2026, 1400

Coursework summary

This worksheet supplements material on the history of computer hardware delivered in week 1’s lectures. Using internet resources, you will perform research to answer the following questions about historical and recent computing devices.

Learning outcomes

On completion of this activity sheet you will:

1. Recognise the contributions of individuals and organisations in the computing field.
2. Explain trends in the history of the computing field.
3. Describe the basic building blocks of computers and their role in the historical development of computer architecture.

Portfolio Task

The portfolio task is a Gradescope online quiz. You will find a link to it in the COMP1860 Minerva module.

Formative Questions

Mechanical Devices

Antikythera Mechanism

1. When does the Antikythera Mechanism date to?
2. What was it used to calculate?
3. (*) Recent research (27 June 2024) has corroborated a hypothesis about the rings in the device. What is this hypothesis and which UK university corroborated it?

Al-Jazari's Castle Clock

1. When does it date to?
2. What was it used to calculate?
3. (*) It has been described as the earliest programmable computer? How could it be programmed?

Charles Babbage's Analytical Engine

1. When does it date to?
2. It has been described as the first general purpose computer. What similarities to modern computers did it have?
3. The Analytical Engine could be programmed from instructions loaded into the machine. What was the method used to do this? What earlier machine was this method inspired by?
4. (*) Who is credited as the first computer programmer? What did this person show the Analytical Engine could calculate by detailing the steps it would undertake to perform this calculation?

Electromechanical Devices

Konrad Zuse's Z3

1. When does it date to?
2. What was the principal component of its design? How many were used?
3. The Z3 was the first binary computer. What word size did it use?
4. (*) Which US mathematician (a seminal figure in computer science) also showed that electrical relays could be used to create Boolean circuits. When was this published?

Electronic Devices

Colossus

Colossus is regarded as the first programmable electronic digital computer.

1. When does it date to?

2. What was it used for?
3. What was the component used to perform Boolean logic and arithmetic operations?
4. Who were its designers?
5. (*) What did the US National Security Agency use Colossus for after WWII?

Manchester Baby

The Manchester Baby was the first stored-program electronic computer.

1. When does it date to?
2. What arithmetic operations could it perform in hardware?
3. What did the binary instruction 000 do?
4. (*) What was the name of the commercial computer that was a direct successor of the Baby?

Transistor Computers

1. At which UK University was the first experimental transistor computer devised?
2. What was the name of MIT's 1956 experimental transistor computer? What company did the designers go on to form?
3. What was the name of the first commercial fully transistorised calculator?

Integrated Circuits

The creation of metal-oxide semiconductor field-effect transistors (MOSFET) in 1960 at Bell Laboratories gave rise to very large-scale integration (VLSI). General Microelectronic created the first commercial MOSFET integrated circuit (IC).

1. When was it released?
2. How many transistors were on the chip?

Mainframe Computers

Mainframe computers were the earliest form of commercial computer. These machines were large and the components typically filled a room. The IBM System/360 was a popular mainframe computer.

1. When was it first released?
2. What was its word size?
3. How many registers did its processor have?

Minicomputers

Minicomputers were much smaller than their mainframe predecessors. The entirety of the hardware could now be contained in a cabinet.

1. Which well-known operating system was first designed to run on the PDP-7 minicomputer?
2. (*) The VAX minicomputer used its own native operating system. What was it called?

Personal Computers

In the 1970s, computing devices could be built reliably and cheaply enough that they were now affordable for individuals. The first commercially successful personal computer was released in 1974.

1. Which company produced it and what was it called?
2. What CPU did it use?
3. (*) Which well-known software company produced the programming language for it?

Modern CPUs

x86

x86 describes the architecture of many CPUs today.

1. What was the original processor of this family?
2. When was it released?
3. (*) Two instruction set extensions are provided by x86 chip manufacturers to support hardware-assisted virtualization. What are they called?

ARM

ARM is another popular CPU architecture.

1. When was ARM introduced?
2. Is its design CISC or RISC? ?
3. One of its principal designers is a Professor at a UK university. What is his name and which University is he associated with?
4. (*) Give the name of a recent ARM based CPU or SoC (system on chip)?

CPUs

1. (*) For the following recent processors:
 - Categorise each as a CISC or RISC based architecture.
 - Give the number of transistors on the chip (reported or estimate).
 - (a) Intel Core i9-13900K
 - (b) Apple M2 Max
 - (c) AMD Epyc Bergamo

Computer Architecture

1. Explain what is meant by *architecture* and *organisation* with respect to computer hardware?
2. List the common functions of computers?
3. Explain why a *hierarchical structure* is used to design computers?
4. Describe the four components of the *von Neumann* architecture?
5. Explain how the *von Neumann* and *Harvard* architectures differ
6. Describe and explain the fundamental building blocks components of a computer?
7. Describe and explain the fundamental electronic building block of a computer?
8. Define Moore's Law?
9. Describe what an instruction set architecture defines?
10. Explain the difference between Complex Instruction Set Computer (CISC) and Reduced Instruction Set Computer (RISC)?